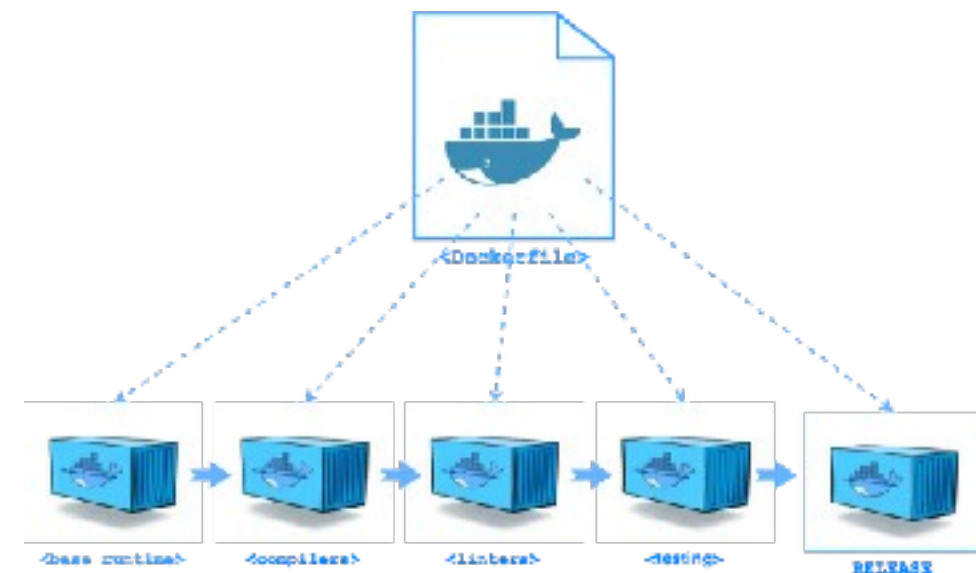
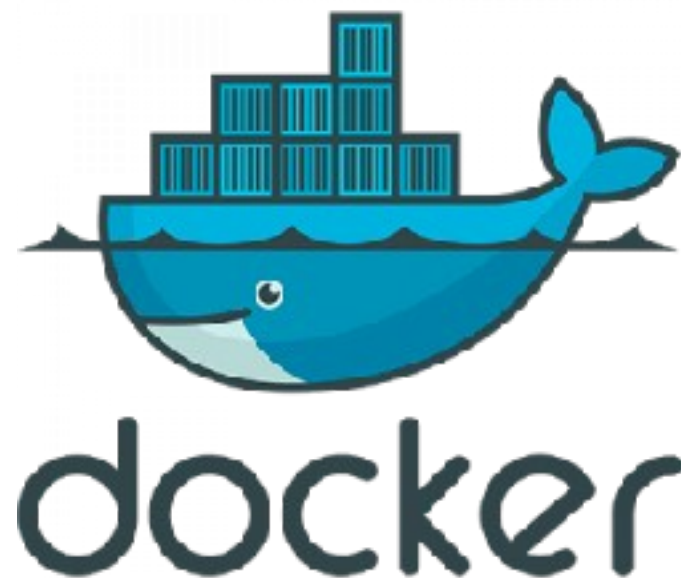


Docker multi-stage build

Using Docker multi-stage build
to create advanced build pipelines



About me...

- Chief of Research [@codefresh.io](https://codefresh.io)
- github.com/alexei-led/pumba
- #docker, #golang, #aws
- medium.com/@alexeiled
- [@alexeiled](https://twitter.com/alexeiled)



Docker build

- **Dockerfile** - imperative DSL that defines build commands
- Each `docker build` command generates ONE image layer
- Complete **docker build** execution generates ONE Docker image

28 lines (21 slot)	855 Bytes	Sourcegraph	Raw	Blame
1	FROM debian:jessie			
2				
3	MAINTAINER NGINX Docker Maintainers "docker-maint@nginx.com"			
4				
5	ENV NGINX_VERSION 1.10.1-1~jessie			
6				
7	RUN apt-key adv --keyserver hkp://pgp.mit.edu:80 --recv-keys 573BFD693D8FBC641079A6AB8F53D827309BFE2 \			
8	&& echo "deb http://nginx.org/packages/debian/ jessie nginx" >> /etc/apt/sources.list \			
9	&& apt-get update \			
10	&& apt-get install --no-install-recommends --no-install-suggests -y \			
11	ca-certificates \			
12	nginx=\${NGINX_VERSION} \			
13	nginx-module-xslt \			
14	nginx-module-geoip \			
15	nginx-module-image-filter \			
16	nginx-module-perl \			
17	nginx-module-njs \			
18	gettext-base \			
19	&& rm -rf /var/lib/apt/lists/*			
20				
21	# forward request and error logs to docker log collector			
22	RUN ln -sf /dev/stdout /var/log/nginx/access.log \			
23	&& ln -sf /dev/stderr /var/log/nginx/error.log			
24				
25	EXPOSE 80 443			
26				
27	CMD ["nginx", "-g", "daemon off;"]			

nginx Dockerfile

21 lines (14 slot)	497 Bytes
1	FROM nginx
2	
3	MAINTAINER Jan Löffler "mail@jlsoft.de"
4	
5	# Upgrade everything
6	RUN apt-get update && apt-get upgrade -y
7	
8	# nginx site conf
9	COPY config/nginx.conf /etc/nginx/nginx.conf
10	COPY config/nginx-site.conf /etc/nginx/sites-available/default
11	
12	# Add website
13	COPY /www /usr/share/nginx/html
14	
15	# forward request and error logs to docker log collector
16	RUN ln -sf /dev/stdout /var/log/nginx/access.log \
17	&& ln -sf /dev/stderr /var/log/nginx/error.log
18	
19	EXPOSE 80 443
20	
21	CMD ["nginx", "-g", "daemon off;"]

html-demo Dockerfile

janloeffler/html-demo Image Layers

16	68.2 MB	nginx 1.11.5	What's this?
	49.0 MB	ADD file:41ea5187c50116884c38d9ec51d920d79cfaeb2a61c...	
		CMD ["/bin/bash"]	
		MAINTAINER NGINX Docker Maintainers "docker-maint@h...	
		ENV NGINX_VERSION=1.11.5-1~jessie	
	19.2 MB	RUN apt-key adv --keyserver hkp://pgp.mit.edu:80 --r...	
	196 bytes	RUN ln -sf /dev/stdout /var/log/nginx/access.log &&...	
		EXPOSE 443/tcp 80/tcp	
		CMD ["nginx" "-g" "daemon off;"]	
		MAINTAINER Jan Löffler "mail@hiddenl"	
	11.9 MB	RUN apt-get update && apt-get upgrade -y	
	815 bytes	COPY file:42f9c1d623748f6ed7b8d73c89da66562765ab221cc2cd655b6223d82a...	
	459 bytes	COPY file:d8783c0369225579c00c032fe9b398ef65ee19a2eb3246163d7c96c17c...	
	229.8 kB	COPY dir:ff268d7a0ecd4997945422e8d10fd70fb16eaa5ba1a320ca7ea4e95cf69...	
	195 bytes	RUN ln -sf /dev/stdout /var/log/nginx/access.log && ln -sf /dev/std...	
		EXPOSE 443/tcp 80/tcp	
	32 bytes	CMD ["nginx" "-g" "daemon off;"]	

Problem with Docker Build

Image we want

runtime
configuration
application

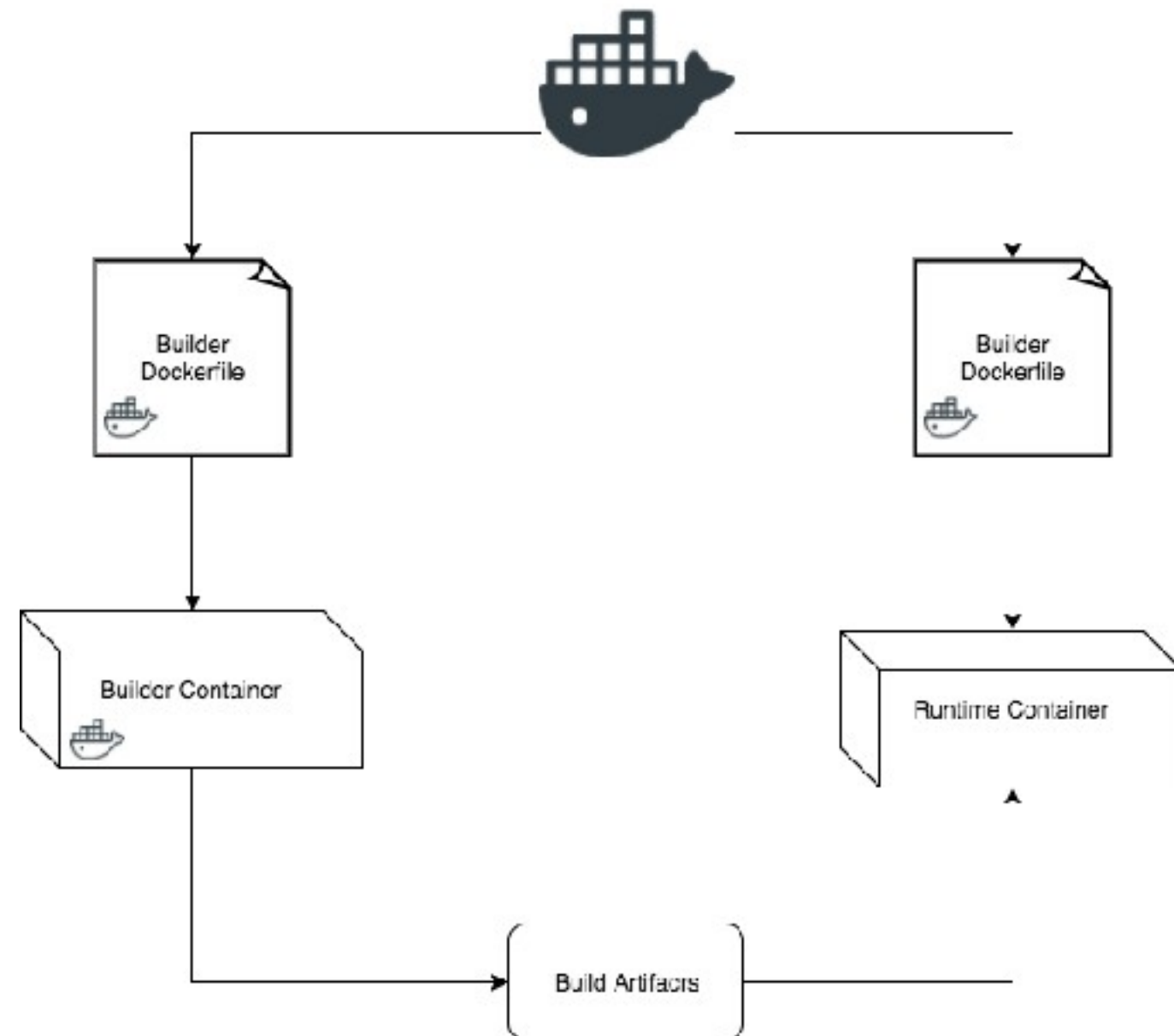
x (4..10)

Image we build

compilers, debuggers, ...
linters, tests, profilers, ...
code, build and test logs, ...
runtime
configuration
application

Docker Builder Pattern

- 2 Dockerfiles
 - 1st for build tools
 - 2nd for runtime



Drawbacks

- 2+ Dockerfiles
- Orchestration

Docker multi-stage build

✓ Benefits

- ✓ One Dockerfile
- ✓ One syntax to learn
- ✓ Same Build
 - ✓ Local and CI
- ✓ Create multiple stages

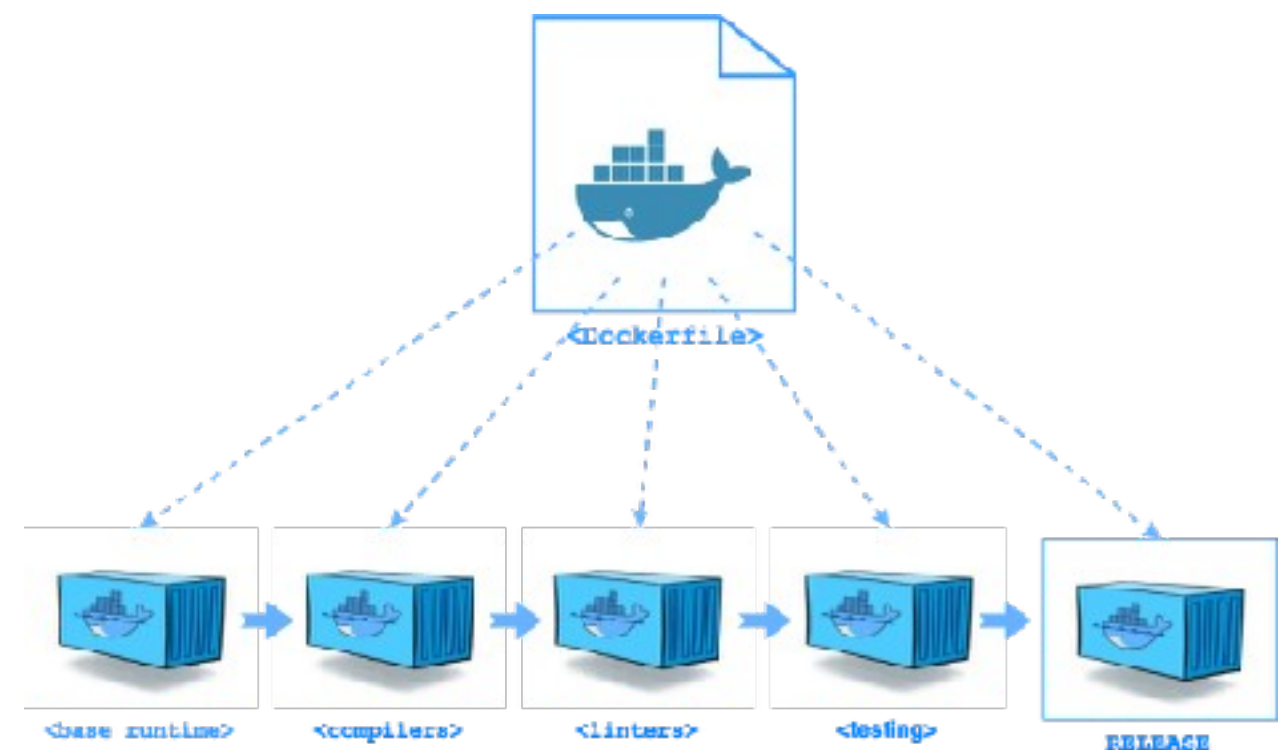
Build smaller images with
Multi-stage builds

First stage:
complete build
environment

Second stage:
minimal runtime
environment ❤️



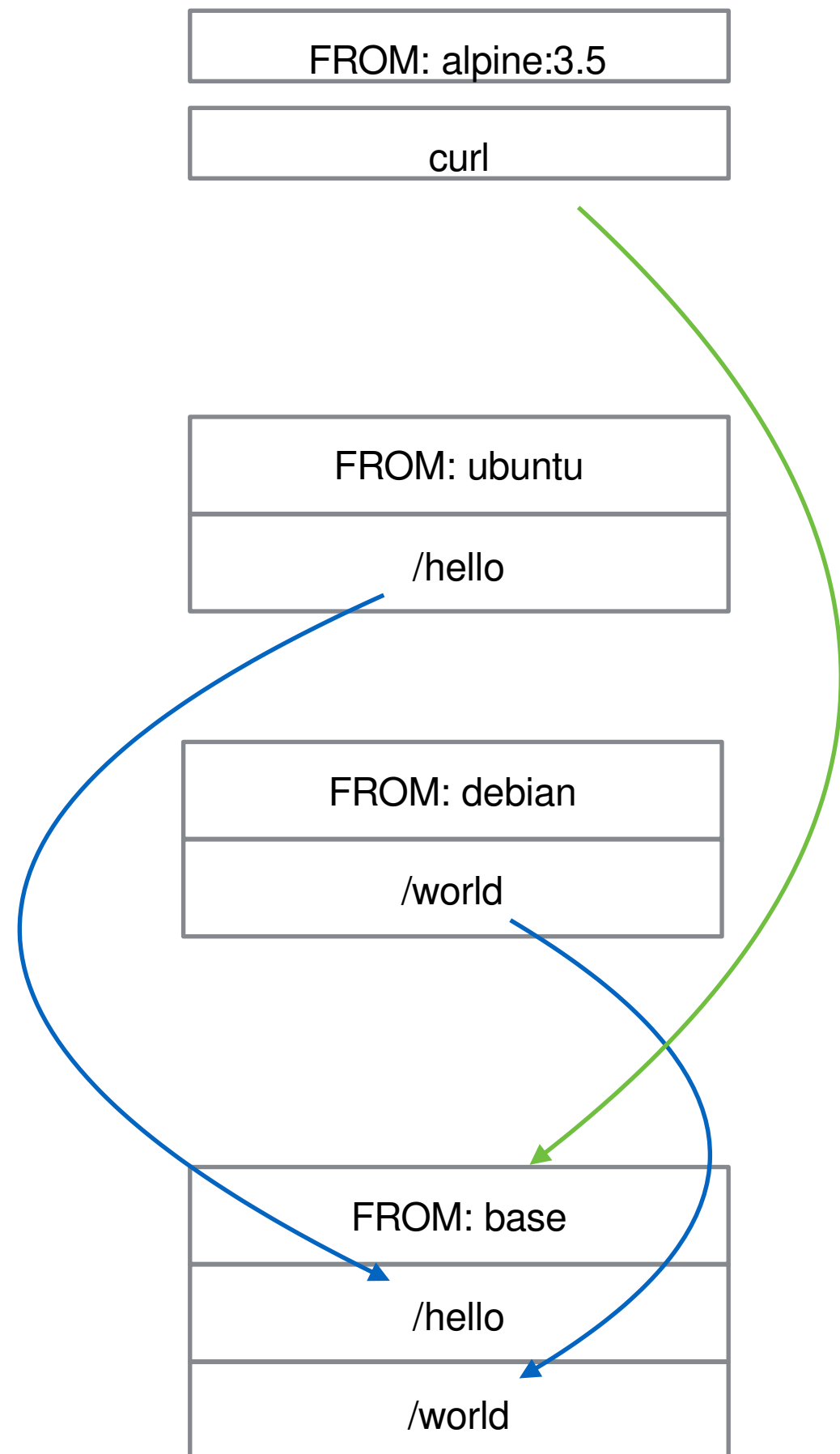
One Dockerfile, one build





Dockerfile ✕

```
1  # Base Image
2  FROM alpine:3.5 AS base
3  RUN apk add --no-cache curl
4
5  # Second Image
6  FROM debian AS second
7  RUN echo hello > /hello
8  LABEL image=second
9
10 # Third Image
11 FROM ubuntu AS third
12 RUN echo world > /world
13 LABEL image=third
14
15 # FINAL Image
16 FROM base
17 # Copy files from other images
18 COPY --from=second /hello /hello
19 COPY --from=third /world /world
20
21
```



Node.js multi-stage build

<https://github.com/codefreshdemo/demochat/tree/multistage>

Let's Chat

	Builder (Alpine)	Runtime (Alpine)
	Node.js	Node.js
Content	App Modules Yarn (package) Mocha (testing) Eslint (linter) Newman (REST testing) Gulp (automation) nsp (security scanner) Istanbul (coverage) Python gcc, g++, krb5-dev, ...	App Modules Yarn (package)
Size	568 MB	112 MB

Go multi-stage build

<https://github.com/gaia-adm/pumba/tree/multistage>



	Builder (Alpine)	Runtime (Alpine)
Content	go 1.8 glide (package) cover (coverage) go test (testing) git github-release (release) goveralls (coveralls.io) Bats (testing) Bash curl, openssl Docker CLI	pumba
Size	427 MB	15 MB

Docker multi-stage with codefresh.io

Summary

Dockerfile

Logs

Layers



gaia-adm/pumba

multistage/bb017b8

pumba

Download log

+ Initializing Process

9 seconds

✓

+ Cloning main repository...

2 seconds

✓

- multistage-build

53 seconds

✓

Removing intermediate container 31c4c10c0d91

Successfully built aa39ddbe2c93

Successfully tagged gaiaadm/pumba:dff7693a11fb613f60ae3d077a2e937c9ff1f1e6

Successfully built image: gaiaadm/pumba:dff7693a11fb613f60ae3d077a2e937c9ff1f1e6

The push refers to a repository [gcr.io/codefresh-production/gaiaadm---pumba]

Layer 'e5907091b4f3' already exists

Layer '572a8c2e054c' already exists

Layer 'c154057080f4' already exists

Layer '33e9e0cf8388' successfully pushed

Layer '3f734bc277cb' successfully pushed

Layer '14ec24a95853' successfully pushed

dff7693a11fb613f60ae3d077a2e937c9ff1f1e6: digest: sha256:e2c4ccc300fc369337c1b35e1799b233af327f440d9c139135657cc8ea283302 size: 1571

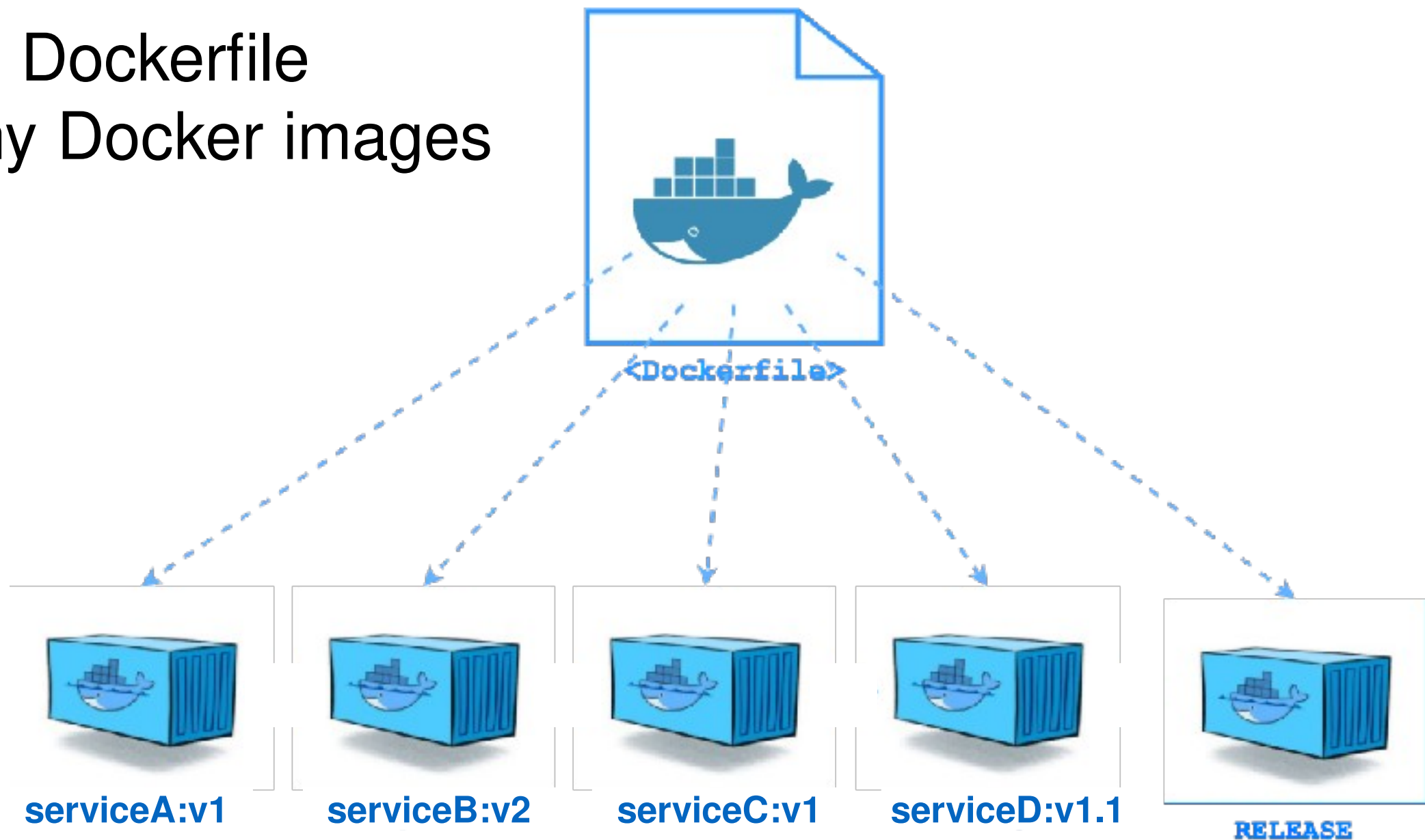
☐ Autoscroll with output

 Copy to clipboard

Completed in: ⌚ 1 min 4 sec

Multi-build Use Case

- ✓ One Dockerfile
- ✓ Many Docker images



```
$ label = "LABEL name=serviceA"
```

```
$ tag = "serviceA:v1"
```

```
$ docker tag $(docker history 224f4fee5ec1 | grep $label | awk '{print $1}') $tag
```

Docker Builder Pattern

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https://codefresh.io/blog/node_docker_multistage/